Device Interaction Framework: Type-Safe Network Device Automation in Rust

Multi-Backend Transport, Fluent Verification, and Declarative Test Suites

Simon Knight

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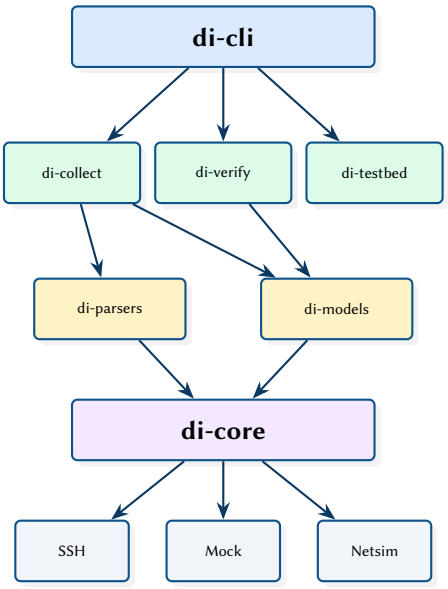

Abstract. This report presents a Rust framework for network device interaction and automated testing, inspired by Cisco's pyATS but designed for type safety and performance. The framework provides multi-backend transport (SSH, Mock, Netsim), strongly-typed domain models for interfaces, routes, and ping results, a fluent verification API with domain-focused error reporting, and declarative YAML test suites that define network state assertions without code. The system is organised as seven Rust crates: core transport, parsers, models, collection, verification, testbed management, and CLI.

Version	Date	Changes
0.1.0	March 2026	Initial architecture report

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List of Design Decisions & Rules

1 Introduction and Motivation

Network automation frameworks like pyATS [1] provide powerful device interaction capabilities but are constrained by Python's runtime performance, lack of compile-time type checking, and the GIL's impact on concurrent device access. The Device Interaction Framework reimplements these patterns in Rust, providing compile-time safety, zero-cost abstractions, and true parallelism for multi-device operations.

: Type-Safe Domain Models

All parsed device output is represented as strongly-typed Rust structs (not dictionaries or JSON). Interface status, routing table entries, and ping results are modelled with enums and newtypes, catching invalid states at compile time rather than runtime.

2 System Architecture

The framework is organised as seven crates with a layered dependency structure:

Table 1. Crate responsibilities.

Crate	Layer	Role
`di-core`	Transport	Connection management, backend trait
`di-parsers`	Parsing	CLI output parsing (TextFSM-style)
`di-models`	Domain	Interface, Route, Ping models
`di-collect`	Collection	High-level device interaction API
`di-verify`	Verification	Fluent assertion engine
`di-testbed`	Inventory	Testbed YAML, credentials
`di-cli`	Application	Unified CLI binary

2.1 Transport Backends

The `di-core` crate defines a `Transport` trait that abstracts device communication:

```
#[async_trait]
pub trait Transport: Send + Sync {
    async fn connect(&mut self) -> Result<()>;
    async fn execute(&mut self, command: &str) -> Result<String>;
    async fn disconnect(&mut self) -> Result<()>;
    fn is_connected(&self) -> bool;
}
```

Three backends implement this trait:

- **SSH** — Production backend using `russh` for async SSH2 connections with prompt detection and command timing.
- **Mock** — Development/CI backend that replays pre-recorded command–response pairs from YAML fixtures.
- **Netsim** — Integration backend that connects to a network simulator's management API.

Design Decision 1: Mock Backend for CI

The Mock transport enables full integration testing without network hardware. Test fixtures capture real device output once, then replay deterministically. This allows the verification suite to run in CI with sub-second execution times.

3 Verification API

The `di-verify` crate provides a fluent assertion API inspired by property-based testing frameworks:

```
verify(&device)
    .interfaces()
    .named("GigabitEthernet0/0")
    .is_up()
    .has_ip("10.0.0.1/30")
    .and()
    .routes()
    .to("192.168.0.0/16")
    .via("10.0.0.2")
    .exists()
    .run()
    .await?;
```

Failed assertions produce domain-specific error messages:

```
FAIL: Interface GigabitEthernet0/0
  Expected: status = Up
  Actual:   status = Down (administratively)
  Device:   spine-1 (10.1.1.1)
```

3.1 Declarative Test Suites

For code-free testing, `di-verify` supports YAML test suites:

```yaml
suite: connectivity-check
tests:
  - name: spine uplinks are up
    device: spine-1
    checks:
      - type: interface
        name: "Ethernet1"
        status: up
      - type: route
        prefix: "0.0.0.0/0"
        next_hop: "10.0.0.1"
```

Insight: *The YAML suite format enables network operators who do not write Rust to define automated checks. The CLI loads and executes suites against the testbed inventory, reporting results in JUnit XML format for CI integration.*

4 Design Summary

Table 2. Key design decisions.

Decision	Rationale
Seven-crate workspace	Clean dependency boundaries; transport layer does not depend on domain models
Async transport trait	Enables concurrent multi-device operations without thread-per-connection
Mock backend	Deterministic CI testing without hardware; sub-second test execution
Fluent verification API	Readable assertions with domain-specific error messages
YAML test suites	Code-free testing for network operators

References

[1] Cisco, *pyATS: Python automated test systems*, 2024. [Online]. Available: https://developer.cisco.com/pyats/